

Indian Statistical Institute
M.Tech. (CS), Final of Second Semester Examination, 2025-26
Database Management Systems

Full Marks: 50

Date: 25-04-2026

Time: 2.5 Hours

Answer any *five* of the following questions

5 × 10 = 50

1. (a) Suppose the details of the drivers running different cars within a company are given in the following schema. Assume that all entries in the relations are in lowercase.

DRIVERS = $\langle \underline{\text{driverid}} : \text{string}, \text{drivename} : \text{string}, \text{rating} : \text{integer}, \text{age} : \text{integer} \rangle$

CARS = $\langle \underline{\text{carid}} : \text{string}, \text{carname} : \text{string}, \text{color} : \text{string} \rangle$

DRIVES = $\langle \underline{\text{driverid}} : \text{string}, \underline{\text{carid}} : \text{string}, \underline{\text{day}} : \text{date} \rangle$

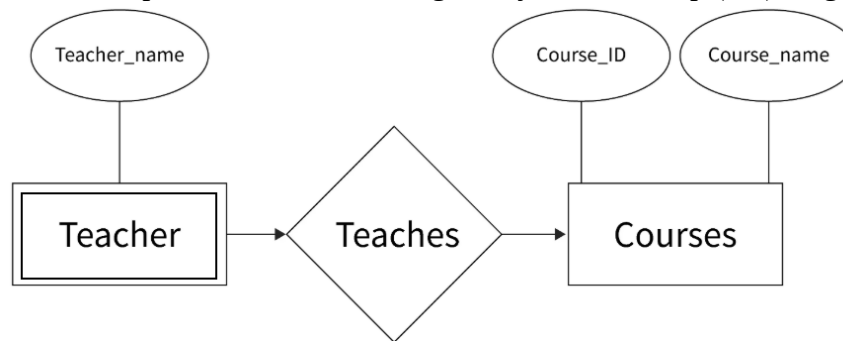
What will be returned by the following relational algebra expressions?

(i) $\pi_{\text{color}}(\sigma_{(\text{carname} \geq 'm') \wedge (\text{carname} < 'n')}(\text{CARS}))$

(ii) $\pi_{\text{drivename}}(\sigma_{\text{age} > 60}(\text{DRIVERS}) \bowtie \text{DRIVES})$

(iii) $\pi_{\text{drivename}}(\sigma_{\text{color} = 'red'}(\text{CARS}) \bowtie \text{DRIVERS} \bowtie \text{DRIVES})$

- (b) Identify the errors present in the following Entity-Relationship (ER) diagram.



(2+2+2)+4

2. (a) Consider a relation in which all attributes are atomic and no non-trivial functional dependencies exist. What is the highest normal form that the relation satisfies? Justify your answer.
- (b) Let there be a relational schema $R = \langle A, B, C, D, E, F \rangle$ holding the set of functional dependencies $FD = \{A \rightarrow BC, B \rightarrow D, E \rightarrow F\}$.
- (i) Suggest a lossless decomposition of this into BCNF. If it is not possible then argue with proper justifications.
- (ii) Suggest a dependency preserving decomposition of this into BCNF. If it is not possible then argue with proper justifications.

4+(3+3)

3. (a) Consider the relations $R1 = \langle \underline{X}, Y, Z \rangle$ and $R2 = \langle \underline{X}, W, V \rangle$ having 120 and 720 tuples, respectively. Note that the primary keys in the respective relations are underlined

therein. The count of distinct values of Y, which starts from 0, in R1 is 6. The range of values of W in R2 is [10, 160]. Estimate the size of the following query considering the attribute values are integers and uniformly distributed.

$$\sigma_{Y \neq 0}(R1) \bowtie \sigma_{W < 50}(R2)$$

- (b) The following information pertaining to the relations $R1 = \langle \underline{A}, B, C \rangle$, $R2 = \langle \underline{A}, B, C \rangle$ and $R3 = \langle \underline{A}, B, C \rangle$ are provided to you.

- (i) R1, R2 and R3 possess millions of tuples.
- (ii) R1 and R2 share a lot of common tuples.
- (iii) R2 and R3 have no tuple in common.

Convert the following relational algebra expression into a better logical plan.

$$\sigma_{\theta}(R1) - (R2 \bowtie R3)$$

6+4

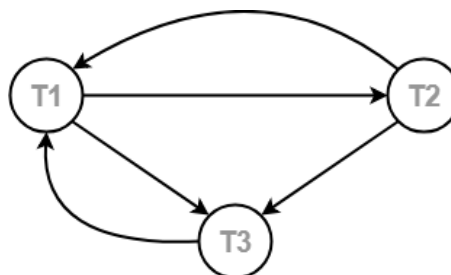
4. (a) Let there be a relation with data items X, Y and Z. The following transactions are required to be scheduled to be processed on the relation. Create a schedule with the following three transactions that must be *view serializable* but not *conflict serializable*. Justify your answer.

T_1
read(X)
$X = X - 100$
write(X)
read(Y)
$Y = Y + 100$
write(Y)
commit

T_2
read(Z)
$Z = Z - 100$
write(Z)
read(Y)
$Y = Y + 100$
write(Y)
Commit

T_3
read(X)
$X = X / 10$
write(X)
read(Z)
$Z = Z + Z * 0.2$
write(Z)
commit

- (b) Identify the serializability order from the precedence graph given below. Comment on whether the schedule corresponding to the precedence graph is *view serializable* or not.



6+(3+1)

5. (a) Suggest potential solutions of the problems with transactions arising from concurrency control that are listed below.
- (i) Starvation of a transaction.
 - (ii) Cascading rollback of transactions.
 - (iii) Uncommitted dependency of a transaction.
 - (iv) Phantom phenomenon.
- (b) Is it possible to implement a *lock table* for concurrency control with stack data structures only? Justify your answer.

(2+2+2+2)+2

6. (a) Consider the following *vector clock*, which uses timestamps for concurrency control, for managing 5 nodes (numbered as 1, 2, 3, 4 and 5).

2	18	8	9	21
16	12	14	15	22
1	3	4	5	24
10	7	13	11	23
17	6	19	20	25

Suppose an update is intended to happen at node 1 at timestamp 26. Is it possible to perform the same? Justify your answer.

- (b) Convert the following table into JSON format.

ID	Name	Age	Gender	Hobby
1	Amar	NULL	M	Geocaching
2	Fatima	24	F	NULL
3	Antony	29	NULL	Phillumeny

6+4

7. Show appropriate examples of the following things.

- (i) A table containing two foreign keys.
- (ii) A table satisfying 4NF but not 5NF.
- (iii) Two tables whose natural join returns one of the tables.
- (iv) An SQL query that is already optimized.
- (v) A schedule that is not view serializable.

2+2+2+2+2
